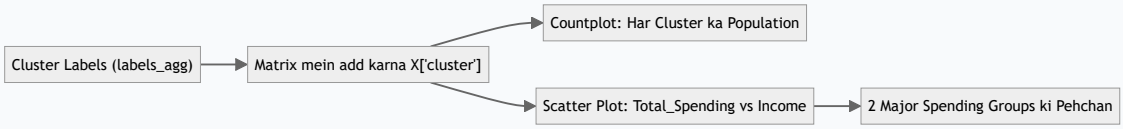
# Submodule 33.11: Cluster Characterization (Income vs Spending Behavioral Patterns) (Hinglish)

## Topic Overview & Grounding Labels

Agglomerative Clustering se mile cluster labels ko original feature space ke sath jodkar har cluster ke physical characteristics aur spending behavior ko analyze kiya gaya.




> [!NOTE] Video lecture recording ke dauran `[01:40] - [02:30]` ke beech audio issue / silence tha. Uske baad explanation smoothly resume hota hai. Code aur explanations notebook se 100% verified hain.

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## 1. Har Cluster ka Population Size

Har cluster mein kitne customers hain:

Reference: smartcart.ipynb (Cells 61 - 63).

```
import seaborn as sns
import matplotlib.pyplot as plt

# Labels ko dataframe mein add karna
X["cluster"] = labels_agg

# Lecture matching palette
pal = ["red", "blue", "yellow", "green"]

# Countplot banana
plt.figure(figsize=(6, 4))
sns.countplot(x=X["cluster"], palette=pal, hue=X["cluster"])
plt.title("Cluster Counts")
plt.show()
```

```
print(X["cluster"].value_counts().sort_index())
```

### Exact Customer Counts ($N = 2,236$):

- **Cluster 0 (Red)**: **905 customers** (40.5% - Sabse bada cluster)
- **Cluster 1 (Blue)**: **534 customers** (23.9%)
- **Cluster 2 (Yellow)**: **444 customers** (19.9%)
- **Cluster 3 (Green)**: **353 customers** (15.8%)

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## 2. Income vs Total Spending Scatter Plot

Income aur Total Spending dataset ke do sabse strongly correlated features ($r = +0.79$) hain.

Reference: smartcart.ipynb (Cell 64).

```
plt.figure(figsize=(8, 6))
sns.scatterplot(
    x=X["Total_Spending"],
    y=X["Income"],
    hue=X["cluster"],
    palette=pal
)
plt.title("Income vs Total Spending by Cluster")
plt.xlabel("Total Spending ($)")
plt.ylabel("Income ($)")
plt.show()
```

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## 3. High-Level Patterns & Core Question

| Cluster | Color | Income Range | Spending Range | High-Level Category | | :--- | :--- | :--- | :--- | :--- | | **Cluster 1** | Blue | High ($\sim 72.8k$) | $High$ (1,237) | **High Spenders** | | **Cluster 3** | Green | High ($\sim 70.7k$) | $High$ (1,190) | **High Spenders** | | **Cluster 0** | Red | Moderate/Low ($\sim 39.7k$) | $Moderate/Low$ (222) | **Budget Shoppers** | | **Cluster 2** | Yellow | Lowest ($\sim 37.0k$) | $Lowest$ (166) | **Budget Shoppers** |

### Interesting Insight:

Customers do distinct pairs mein divide hue:

1. Agar Cluster 1 aur Cluster 3 dono high income aur high spending hain, toh algorithm ne unhe alag-alag clusters mein kyun divide kiya?

2. Agar Cluster 0 aur Cluster 2 dono low income aur low spending hain, toh unhe do alag clusters mein kyun baanta?

Iska jawab demographic features mein chupa hai: **Household structure (`Living_With`) aur ghar mein bacchon ki sankhya (`Total_Children`)**, jise agle module mein detail se dekhenge!